

**CANDIDATE – PLEASE NOTE!**

PRINT your name on the line below and return this booklet with the answer sheet. Failure to do so may result in disqualification.

TESTCODE **01234010****FORM TP 2017016**

JANUARY 2017

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION
MATHEMATICS**

Paper 01 – General Proficiency*1 hour 30 minutes***04 JANUARY 2017 (p.m.)****READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. This test consists of 60 items. You will have 1 hour and 30 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. **A list of formulae is provided on page 2 of this booklet.**
4. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer, and decide which choice is best.
5. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

$2a + 6a =$

- (A) $8a$
(B) $8a^2$
(C) $12a$
(D) $12a^2$

Sample Answer

The best answer to this item is “ $8a$ ”, so (A) has been shaded.

6. If you want to change your answer, erase it completely before you fill in your new choice.
7. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
8. You may do any rough work in this booklet.
9. Figures are not necessarily drawn to scale.
10. Calculators and mathematical tables are NOT allowed for this paper.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

LIST OF FORMULAE

Volume of a prism $V = Ah$ where A is the area of a cross-section and h is the perpendicular length.

Volume of cylinder $V = \pi r^2 h$ where r is the radius of the base and h is the perpendicular height.

Volume of a right pyramid $V = \frac{1}{3} Ah$ where A is the area of the base and h is the perpendicular height.

Circumference $C = 2\pi r$ where r is the radius of the circle.

Arc length $S = \frac{\theta}{360} \times 2\pi r$ where θ is the angle subtended by the arc, measured in degrees.

Area of a circle $A = \pi r^2$ where r is the radius of the circle.

Area of a sector $A = \frac{\theta}{360} \times \pi r^2$ where θ is the angle of the sector, measured in degrees.

Area of trapezium $A = \frac{1}{2} (a + b) h$ where a and b are the lengths of the parallel sides and h is the perpendicular distance between the parallel sides.

Roots of quadratic equations If $ax^2 + bx + c = 0$,

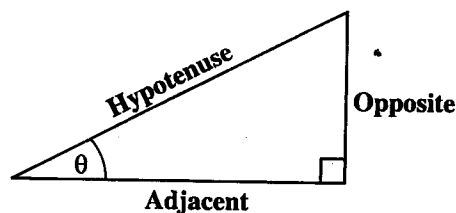
$$\text{then } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Trigonometric ratios

$$\sin \theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite side}}{\text{adjacent side}}$$



Area of triangle

Area of $\Delta = \frac{1}{2} bh$ where b is the length of the base and h is the perpendicular height.

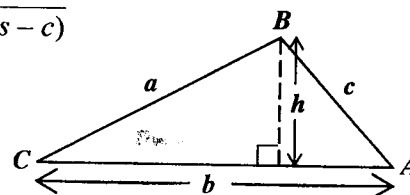
$$\text{Area of } \Delta ABC = \frac{1}{2} ab \sin C$$

$$\text{Area of } \Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{where } s = \frac{a+b+c}{2}$$

Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Cosine rule

$$a^2 = b^2 + c^2 - 2bc \cos A$$

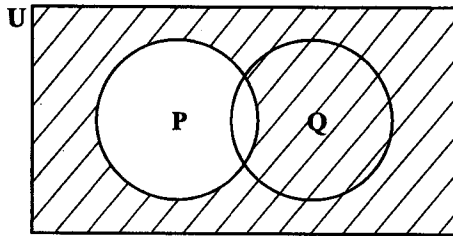
1. $\left(\frac{2}{3}\right)^2$ is equal to
- (A) $\frac{4}{6}$
(B) $\frac{4}{3}$
(C) $\frac{2}{9}$
(D) $\frac{4}{9}$
2. If $4.3 \times 0.37 = 1.591$, then 0.43×370 is
- (A) 1.591
(B) 15.91
(C) 159.1
(D) 1591.0
3. In standard notation, 208.06 is written as
- (A) 2.0806×10^2
(B) 20.806×10^1
(C) 2.0806×10^{-1}
(D) 0.20806×10^3
4. If \$560 is shared in the ratio 2:3:9, the difference between the largest and the smallest shares is
- (A) \$ 80
(B) \$240
(C) \$280
(D) \$360
5. If 60% of a number is 90, what is $\frac{4}{5}$ of the number?
- (A) 54
(B) 72
(C) 120
(D) 150
6. There are 40 students in a class. Girls make up 60% of the class and 25% of them wear glasses. How many girls in the class wear glasses?
- (A) 6
(B) 8
(C) 10
(D) 15
7. If $3n$ is an even number, which of the following is an odd number?
- (A) $3n - 1$
(B) $3n - 2$
(C) $3n + 2$
(D) $3n + 2n$
8. 99×101 has the same value as
- (A) $(99 \times 100) + 1$
(B) $(99 \times 100) - (99 \times 1)$
(C) $(99 \times 100) + (99 \times 1)$
(D) $(99 \times 100)(99 \times 1)$
9. What is the LEAST number of apples that can be shared equally among either 6, 8 or 9 children?
- (A) 36
(B) 48
(C) 54
(D) 72
10. What is the value of the digit 2 in the number 48.621?
- (A) $\frac{2}{100}$
(B) $\frac{2}{10}$
(C) 2
(D) 200



11. If X and Y are two finite sets such that $n(X) = 9$, $n(X \cap Y) = 4$ and $n(X \cup Y) = 15$, then $n(Y)$ is

(A) 5
(B) 6
(C) 10
(D) 12

Item 12 refers to the following Venn diagram.



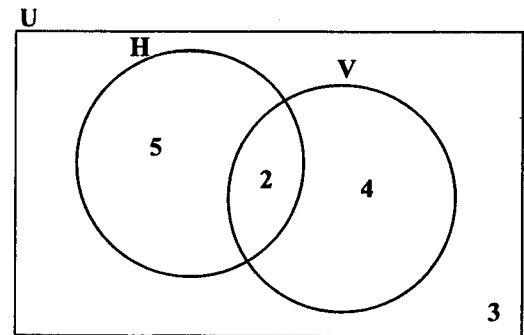
12. In the Venn diagram, the shaded area represents

(A) P'
(B) $(P \cup Q)'$
(C) $Q \cup P'$
(D) $Q \cap P'$

13. Of a class of 32 students, 17 study Music and 20 study Art. What is the LEAST possible number of students who study BOTH Music and Art?

(A) 3
(B) 5
(C) 12
(D) 15

Item 14 refers to the following Venn diagram.



14. In the Venn diagram

$U = \{\text{students who play games}\}$
 $H = \{\text{students who play hockey}\}$
 $V = \{\text{students who play volleyball}\}$

The number of students in each set is indicated. How many students do NOT play volleyball?

(A) 2
(B) 3
(C) 5
(D) 8

15. The cost of a calculator exclusive of sales tax is \$120. Susan bought a calculator and paid sales tax of 10%. How much change would she receive from \$140?

(A) \$ 8
(B) \$12
(C) \$28
(D) \$32

16. At the end of any year a car is worth 5% less than what it was worth at the beginning of the year. If a car was worth \$9500 in December 2016, then its value in January 2017 was

(A) \$ 9 995
(B) \$10 000
(C) \$10 025
(D) \$12 000

17. An article bought for \$125 was sold for \$175. The profit as a percentage of the cost price was

(A) 28.6
(B) 40
(C) 50
(D) 71.4

18. A television set costs \$350 cash. When bought on hire purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30. How much is saved by paying cash?

(A) \$10
(B) \$25
(C) \$40
(D) \$45

19. A company employs 12 gardeners at \$26 per day, and 8 clerks at \$17 per day. What is the mean daily wage, in dollars, of the 20 employees?

(A) \$20.00
(B) \$21.50
(C) \$22.40
(D) \$31.50

Item 20 refers to the following table which shows the rates charged by an insurance company.

House Insurance	25¢ per \$100
Contents Insurance	50¢ per \$100

20. How much will a person pay for insurance, if his house is valued at \$50 000, and the contents at \$10 000 ?

(A) \$175
(B) \$225
(C) \$450
(D) \$500

21. If the simple interest on \$800 for 3 years is \$54, what is the rate of interest per annum?

(A) $\frac{4}{9}\%$
(B) $2\frac{1}{4}\%$
(C) 5%
(D) 44%

22. Tom bought a pen for \$60 and sold it to gain 20% on the cost price. How much money did he gain?

(A) \$12
(B) \$40
(C) \$72
(D) \$80

23. $\frac{4}{5x} + \frac{2}{5x} =$

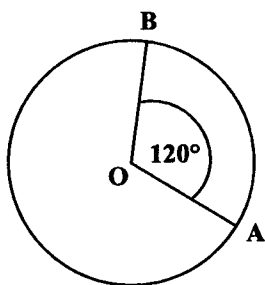
(A) $\frac{6}{25x}$
(B) $\frac{8}{25x}$
(C) $\frac{6}{10x}$
(D) $\frac{6}{5x}$

24. If x is an integer that satisfies the inequality $6 < 2x \leq 8$, then

(A) $3 < x \leq 4$
(B) $3 \leq x < 4$
(C) $-3 < x \leq 4$
(D) $-3 < x < 4$

25. If $x = 4$ and $y = 2$, what is the value of $\frac{x^2 + 3y}{xy}$?
- (A) $1\frac{3}{4}$
(B) $2\frac{1}{2}$
(C) $2\frac{3}{8}$
(D) $2\frac{3}{4}$
26. The expression $-2(x - 4)$ is the same as
- (A) $-2x - 8$
(B) $-2x + 8$
(C) $-2x - 4$
(D) $-2x + 4$
27. If $m * n = \sqrt{m^2 - n^2}$ then $5 * 3 =$
- (A) 2
(B) 4
(C) $\sqrt{8}$
(D) $\sqrt{34}$
28. James saves \$ x each month, but in March he saved \$5 less than twice his usual amount. In March, James saved
- (A) $\$5x$
(B) $\$10x$
(C) $\$x - 5$
(D) $\$2x - 5$
29. If $3(x-1) - 2(x-1) = 7$, then the value of $(x - 1)$ is
- (A) 6
(B) 7
(C) 8
(D) 9
30. $(8a)^2 =$
- (A) $16a$
(B) $64a$
(C) $16a^2$
(D) $64a^2$
31. $2^6 \div 2^2 =$
- (A) 2^3
(B) 2^4
(C) 2^8
(D) 2^{12}
32. Two tonnes, 54 kilograms is the same as
- (A) 56 kg
(B) 254 kg
(C) 2 054 kg
(D) 20 054 kg

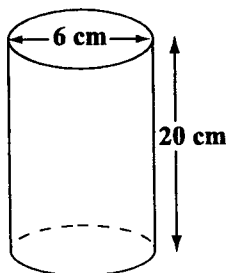
Item 33 refers to the following circle, with centre O.



33. If the circumference of the circle is 15 cm, then the length of the minor arc AB, in centimetres, is

- (A) $\frac{360}{120} \times 15$
 (B) $\frac{120}{360} \times 15$
 (C) $\frac{360}{360-120} \times 15$
 (D) $\frac{360-120}{360} \times 15$

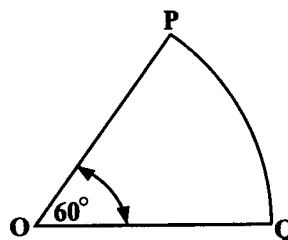
Item 34 refers to the following diagram which shows a cylinder with diameter 6 cm and height 20 cm.



34. The volume, in cm^3 , of the cylinder is

- (A) 180π
 (B) 240π
 (C) 360π
 (D) 720π

Item 35 refers to the following diagram, not drawn to scale, which shows the sector of a circle with centre O.



35. In the diagram, the length of the minor arc, PQ, is 8 cm. What is the circumference of the circle, in cm?

- (A) 16
 (B) 24
 (C) 48
 (D) 64

36. Fifty guests each had 2 glasses of champagne. Each glass held 150 millilitres. How many litres of champagne were used?

- (A) 0.15
 (B) 1.5
 (C) 15
 (D) 150

37. The area of a triangle is 60 cm^2 and its base is 12 cm. What is the perpendicular height, in cm, of the triangle?

- (A) 5
 (B) 10
 (C) 13
 (D) 17

38. The perimeter of a square is 52 cm. What is the area, in cm^2 ?

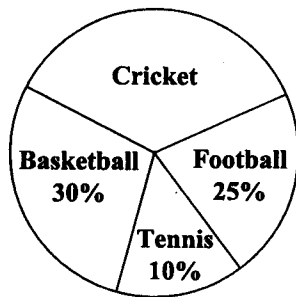
- (A) 39
 (B) 78
 (C) 169
 (D) 196



39. The circumference of a circle is 132 cm. Given that $\pi = \frac{22}{7}$, the radius of the circle, in centimetres, is

(A) 42
(B) 21
(C) $\sqrt{42}$
(D) $\sqrt{21}$

Item 40 refers to the following pie chart which shows the popular games played by 720 students.



40. How many students play cricket?

(A) 35
(B) 120
(C) 252
(D) 300

41. The median of the numbers 1, 3, 4, 5, 5, 9, 9, 9, 9, 10 is

(A) 6.4
(B) 7
(C) 8
(D) 9

42. If the mean of the four numbers 4, 8, x and 12 is 10, then x is

(A) 4
(B) 10
(C) 12
(D) 16

43. When three coins are tossed simultaneously the possible outcomes are {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}, where H represents a head and T represents a tail.

What is the probability of randomly obtaining at LEAST TWO heads when the three coins are tossed simultaneously?

(A) $\frac{1}{4}$
(B) $\frac{3}{8}$
(C) $\frac{1}{2}$
(D) $\frac{2}{3}$

Item 44 refers to the following table which shows the frequency of scores obtained by students in a test.

Scores	2	3	5	6	8	11
Students	12	11	6	3	8	2

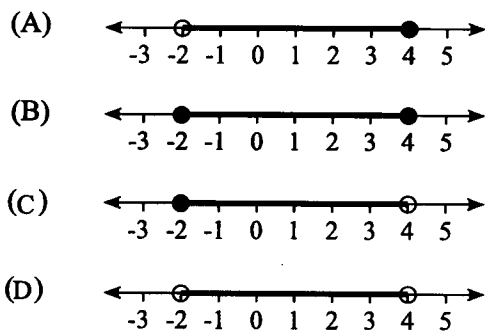
44. The modal score is

(A) 2
(B) 8
(C) 11
(D) 12

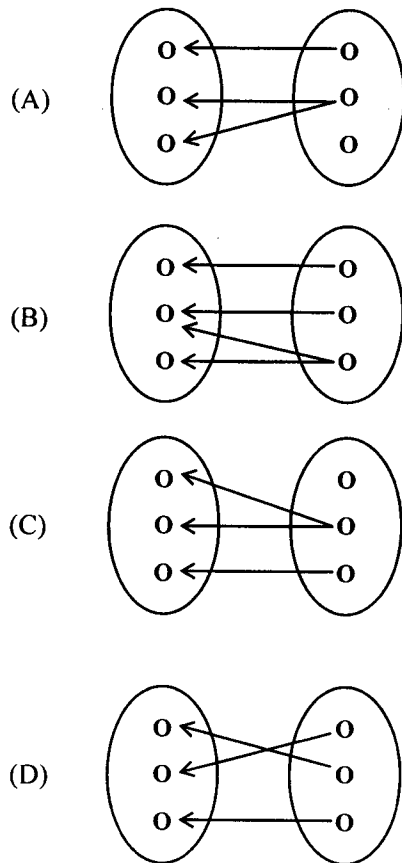
45. Each of the letters of the word 'CHANCE' is written on a slip of paper similar in size and shape. The slips of paper are then placed in a bag and thoroughly shaken. What is the probability of randomly drawing a letter 'C'?

(A) $\frac{1}{6}$
(B) $\frac{1}{5}$
(C) $\frac{1}{3}$
(D) $\frac{2}{3}$

46. Which of the following line graphs represents $\{x : -2 \leq x \leq 4\}$?



47. Which of the following diagrams BEST illustrates a function?



48. The equation of the line which passes through the point $(0, 2)$ and has a gradient of $\frac{1}{3}$ is

- (A) $y = 3x$
 (B) $y = 3x + 2$
 (C) $y = \frac{1}{3}x$
 (D) $y = \frac{1}{3}x + 2$

49. If $f(x) = 2x^2 - 1$, then $f(-3) =$

- (A) -32
 (B) -19
 (C) 17
 (D) 35

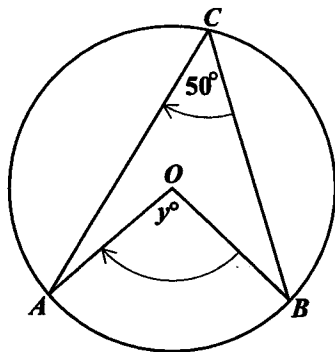
50. Which of the following points lies on the line $y = 2x - 3$?

- (A) $(2, 3)$
 (B) $(4, 1)$
 (C) $(0, -3)$
 (D) $(-2, -1)$

51. The range of $f: x \rightarrow -x^2$ for the domain $\{-2, -1, 0, 1, 2\}$ is

- (A) $\{1, 4\}$
 (B) $\{-4, -1, 0\}$
 (C) $\{-4, -1, 1, 4\}$
 (D) $\{-4, -1, 0, 1, 4\}$

Item 52 refers to the following diagram of a circle.



52. If O is the centre of the circle, then y° is

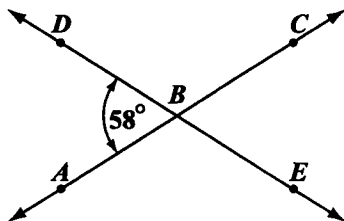
- (A) 25°
- (B) 80°
- (C) 90°
- (D) 100°

53. Under the translation $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ the image of $(3, -5)$ is

- (A) $(1, -2)$
- (B) $(6, 15)$
- (C) $(0, -3)$
- (D) $(5, -8)$

Items 54–55 refer to the following diagram of a pair of intersecting lines.

AC and DE are straight lines intersecting at B . Angle $DBA = 58^\circ$.



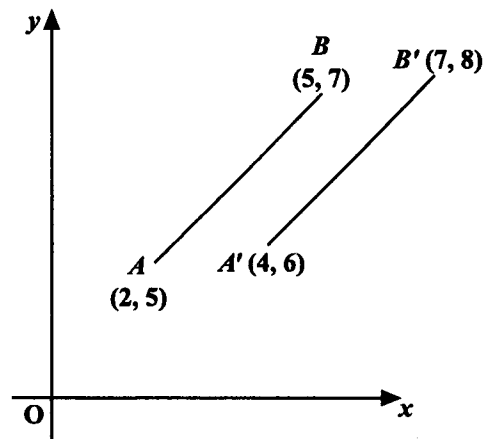
54. The measure of angle ABE is

- (A) 58°
- (B) 122°
- (C) 142°
- (D) 302°

55. Which of the following angles are equal?

- (A) $\angle DBC$ and $\angle CBE$
- (B) $\angle CBE$ and $\angle ABE$
- (C) $\angle ABD$ and $\angle CBD$
- (D) $\angle ABD$ and $\angle CBE$

Item 56 refers to the following diagram of an object and its image after a translation.



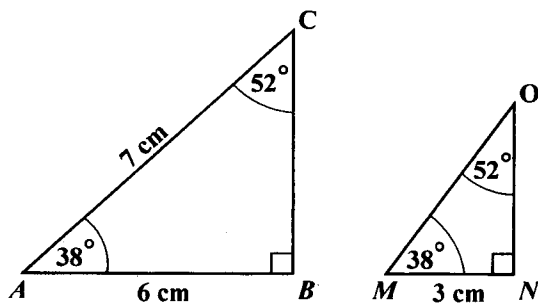
56. In the diagram, the translation by which AB is mapped onto $A'B'$ is represented by

- (A) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- (B) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$
- (C) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$
- (D) $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$

57. In triangle ABC , angle $A = x^\circ$ and angle $B = 2x^\circ$. What is the size of angle C ?

- (A) $(180 - 3x)^\circ$
- (B) 60°
- (C) 30°
- (D) $\left(\frac{180}{3x}\right)^\circ$

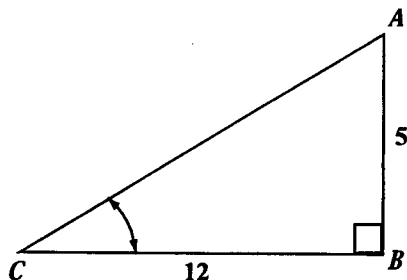
Item 58 refers to the following pair of similar triangles.



58. The length of MO , in centimetres, is

- (A) 3
- (B) 3.5
- (C) 6
- (D) 7

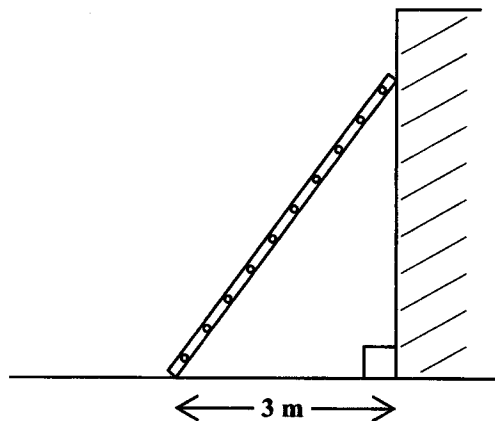
Item 59 refers to the following diagram of a right-angled triangle.



59. From the diagram above, $\sin C$ is

- (A) $\frac{5}{13}$
- (B) $\frac{5}{12}$
- (C) $\frac{12}{13}$
- (D) $\frac{13}{5}$

Item 60 refers to the following diagram which shows a ladder 5 metres long leaning against a vertical wall. The foot of the ladder is 3 metres away from the wall.



60. How far up the wall does the ladder reach?

- (A) 4 m
- (B) 6 m
- (C) 8 m
- (D) 15 m

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

